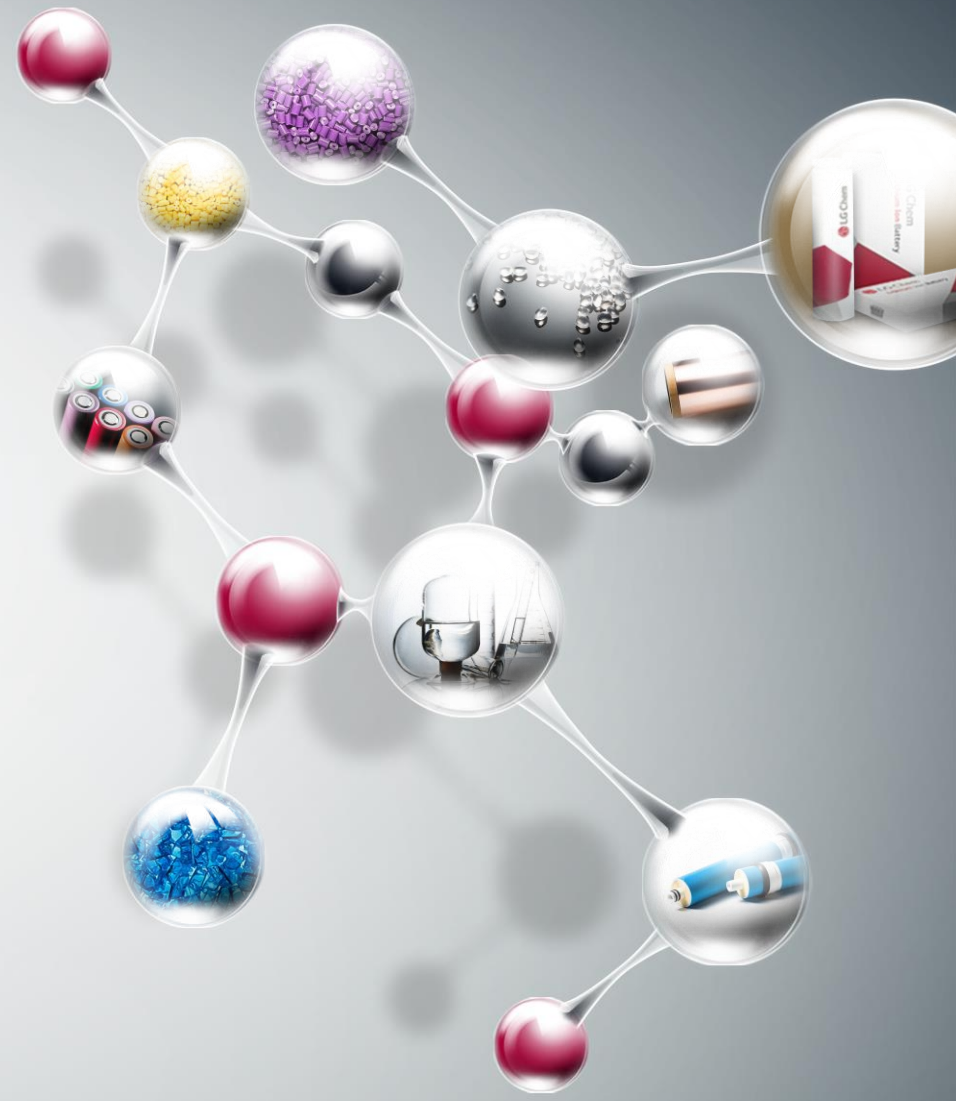
LG Chem

SolutionPartner

LG Chem Impact Modifier

for Engineering Plastics



Contents

Ⅰ LG Chem Overview

Ⅱ LG Chem Impact Modifiers

Ⅲ Impact Modifiers for EP

Ⅰ. LG Chem Overview



LG Chem

Vision

To Be a Global Leader

Growing with customers by providing innovative materials and solutions



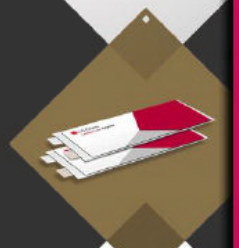
- Customer Value Creation
- Strong Implementation
- Mutual Respect



LG Chem

History









2010's

2017	Merged with LG Life Sciences Co., Ltd
2016	Acquired Dongbu Farm Hannong (Farm Hannong)
2014	NanoH2O (water treatment filter manufacturer) acquired
2010	Started construction of automotive battery plant in Michigan, USA

2000's

2009	Industrial Materials Business spun off (LG Hausys)
2007	Merged with LG Petrochemicals Co., Ltd
2005	Established a polarizer back-end subsidiary in Poland
2004	Established LG Chem (China) Investment Co., Ltd
2004	Completed construction of Ochang Techno Park
2003	Acquired Hyundai Petrochemicals
2001	Split the company (LGCI, LG Chem, LG Household & Healthcare)

1990's

1998	Completed construction Cheongju plant for rechargeable batteries
1995	Completed construction of Tianjin PVC plant in China
1995	Renamed to LG Chem, Ltd.

1980's Earlier

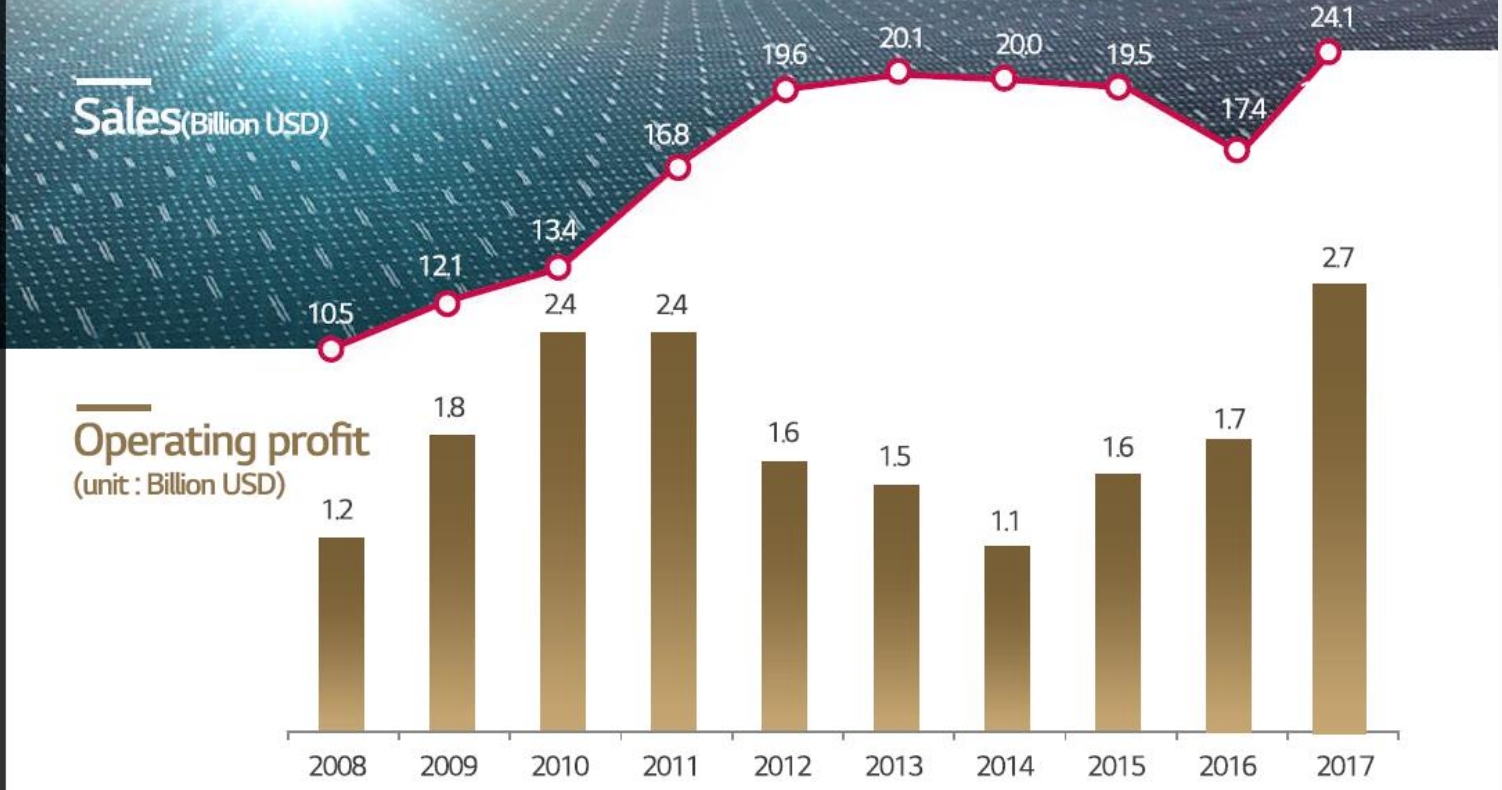
1979	Opened the Daedeok Central R&D Center
1976	Completed the Yeosu PVC Paste Resin Plant
1974	Changed company name to Lucky Corporation
1969	Listed on Korea Stock Exchange
1947	Established as Lucky Chemical Industrial Corporation

LG Chem

Financial Results

Sales in 2017
USD 24.1 Billion

Workforce
29,573 Employees



LG Chem

Domestic Sites



Headquarter/R&D Campus Magok (Est. 1987/Est. 2018)



R&D Campus Gwacheon (Est.2015)



Leadership Center



R&D Campus Daejeon (Est.1979)

Tech Center (Est.1995)



Osong Plant (Est.2009)

Bio Similar, Vaccine



Iksan Plant (3) (Est.1991/1995/2011)

EP, ABS / Pharmaceutical / Battery Materials



Naju Plant (Est.1984)

Octanol, Butanol, Plasticizers



Yeosu Complex (Est.1976)

NCC, PVC, ABS, SAP, PE, AA



Paju Plant (Est.2011)

LCD Glass



Daesan Complex (Est. 2005)

NCC, SSBR, PVC



Ochang Plant 1 (Est. 2005)

Rechargeable Batteries

Polarizer, Stripper



Cheongju Complex (Est.1980)

OLED Material, Photoresist,

Cathode Material, R0 membrane



Gimcheon Plant(Est.2008)

SAP



Ulsan Plant (Est.1974)

Plasticizers

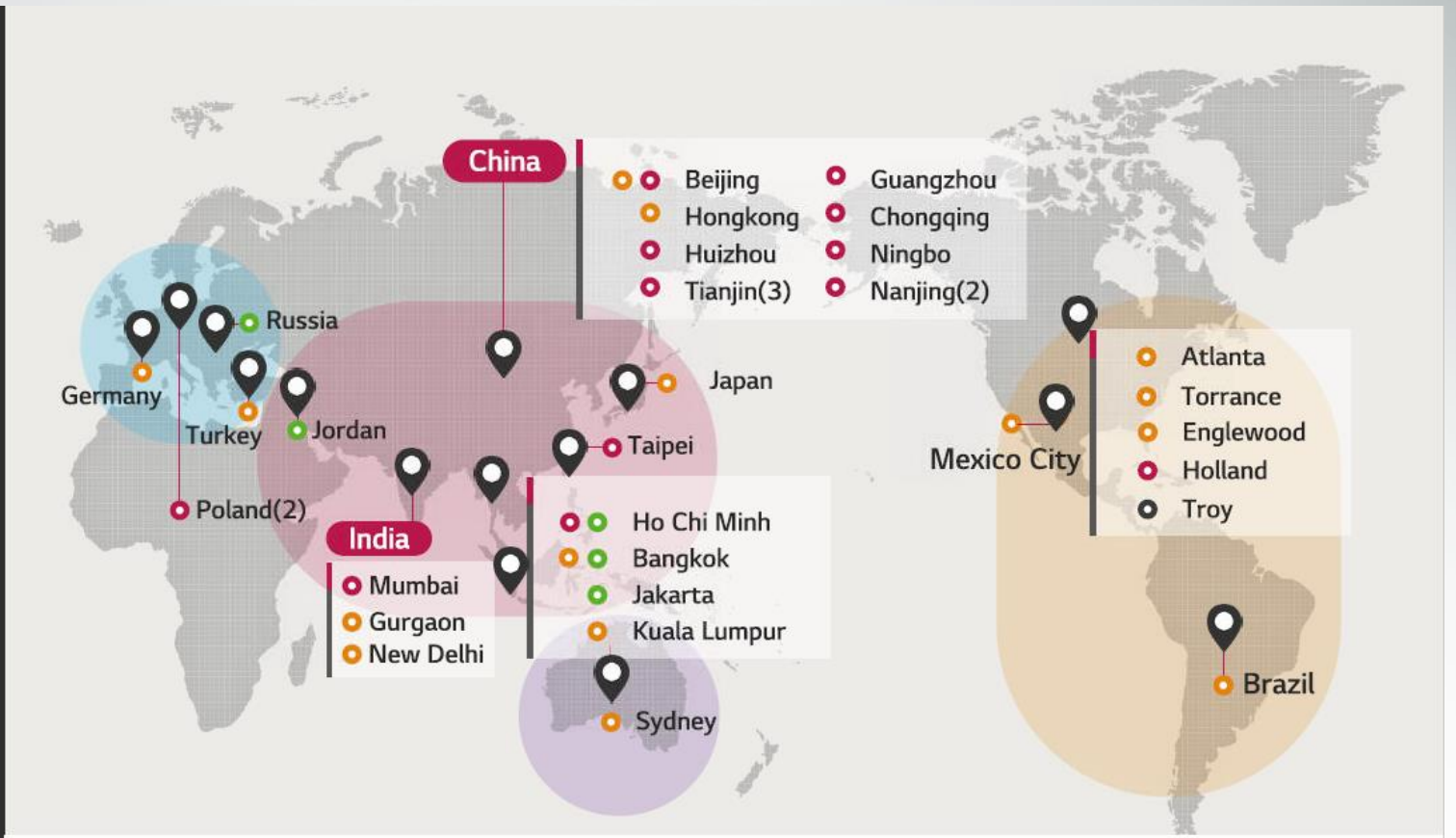


Onsan (Est.1979)

Fine Chemical

LG Chem

Overseas Sites



Asia

- Beijing (Est.2004) - Polarizer
- Tianjin (Est.2004) - EP
  (Est.2005) - PVC,VCM,EDC
  (Est.2009) - SBS
- Guangzhou (Est.2002) - EP
- Chongqing (Est.2015) - EP
- Ningbo (Est.1996) - ABS, SBL, EP
- Nanjing (Est.2003) – Mobile Battery, Polarizer
  (Est.2014) – Automotive Battery
  (Est. 2017) – ESS Battery
- Huizhou (Est.2009) – ABS
- Hongkong
- Taiwan (Est.2004)-Polarizer
- Tokyo
- India (Est1996)-PS, EPS
- Haiphong (Est.2017)-Polarizer
- Ho Chi Minh (Est.1995)-Plasticizers
- Bangkok
- Jakarta
- Kuala Lumpur
- Amman

Australia

- Sydney

America

- Atlanta
- Torrance
- Troy
- Holland (Est.2000) - Automotive Battery
- Sao Paulo
- Mexico City

Europe

- Wroclaw (Est.2005)-Polarizer, EP
  (Est.2016)-Automotive Battery
- Moscow
- Frankfurt
- Istanbul

Manufacturing Subsidiaries (18) | Marketing Subsidiaries (15) | Representative (5) | R&D Center (1)

LG Chem

Business Area



Basic Materials & Chemicals	Energy Solution	IT & Electronics Materials	Advanced Materials	Life Sciences
				
▪ NCC ▪ PolyOlefins ▪ PVC/Plasticizers ▪ ABS ▪ Engineering Plastics ▪ Acrylates/SAP ▪ Rubber/Special Polymers	▪ IT&New Application Battery ▪ Automotive Battery ▪ ESS Battery	▪ Optical Materials ▪ Advanced Functional Materials ▪ LCD Glass Substrates ▪ RO Membrane	▪ Display Materials ▪ Battery Materials	▪ Primary Drug ▪ Specialty Drug ▪ Vaccines

Impact Modifier

II . LG Chem Impact Modifiers



History of LG Chem Impact Modifier



Seoul : Headquarter



Research Park
Tech Center (Est.1995)



Yeosu Complex
(Est.1976)
NCC, PVC, ABS, SAP, PE, AA, MBS, Latex, SBS

- AIM : Impact Modifier based on Acryl rubber
- MBS : Impact Modifier based on Butadiene rubber
- EM : Impact Modifier for Engineering Plastic
- PA : Processing Aid

History

1982 Developed Impact Modifier for PVC
1985 Started Commercial Production of AIM
1990 Started Commercial Production of MBS & PA
2005 Started Commercial Production of EM
2014 Released PA932 for Foamed PVC
2015 Released PA912 for Rigid PVC
2016 Released PA950 for Foamed PVC
Released EM520 for Engineering plastics
2017 Released EM530 for Engineering plastics

Capacity



Capacity : 58,000MT/yr

7 grades of Impact Modifier

4 grades of Processing Aid

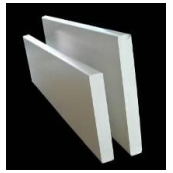
Impact Modifier - MBS, AIM, EM Grade

LG Chem impact modifiers provide excellent impact strength and processability with optimized core-shell structure.

Category		Grade	Features	Application
Impact Modifier based on Butadiene rubber	Transparent	MB872	Excellent transparency Good impact strength	- PVC Applications : Sheet, film, bottle, pipe/fitting, etc 
		MB885	Excellent impact strength Good transparency	
	Opaque	MB838A	More excellent impact strength	
Impact Modifier based on Acryl rubber		IM812	Excellent impact strength & gloss	- PVC Applications : Window profile, siding, sheet, pipe/fitting, etc 
		IM810	Excellent processability Good impact strength	
Impact Modifier for Engineering Plastics (based on Butadiene rubber)		EM520	Excellent impact strength, hydrolytic stability and TVOCs	- PC and PC Alloy with GF/MF reinforced or flame retardant applied. (PC, PC/ABS, PC/SAN, PC/PBT, PC/PET etc) : Electric and Electronic devices, in/exterior of Automobile, etc 
		EM530	Excellent impact strength, melt flow and thermal stability	

Processing Aid

LG Chem processing aids provide excellent mechanical properties as well as excellent fusion properties in PVC application.

Grade	Molecular weight	Features	Application
PA912	Medium	General Purpose PA Excellent processability and dimensional stability	- PVC Applications : Sheet, film, window profile, package, etc   
PA828	High	Excellent processability and dispersibility	
PA932	Very high	Excellent foamability with high melt strength and melt elasticity	- PVC Applications : Foamed product, etc   
PA950	Ultra high	Excellent foamability with high melt strength and melt elasticity	

III. Impact Modifiers for EP

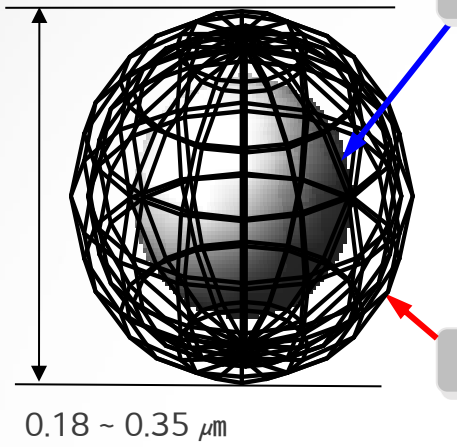
1. Introduction to **EM520 & EM530**
2. Technical Data of **EM520**
3. Technical Data of **EM530**



Toughening Mechanism

LG impact modifier(EM520/EM530) have well-designed core-shell structure. The core-shell structure of the impact modifier is well dispersed in the matrix and interferes with the crack propagation.

Structures of EM520 and EM530



0.18 ~ 0.35 ㎛

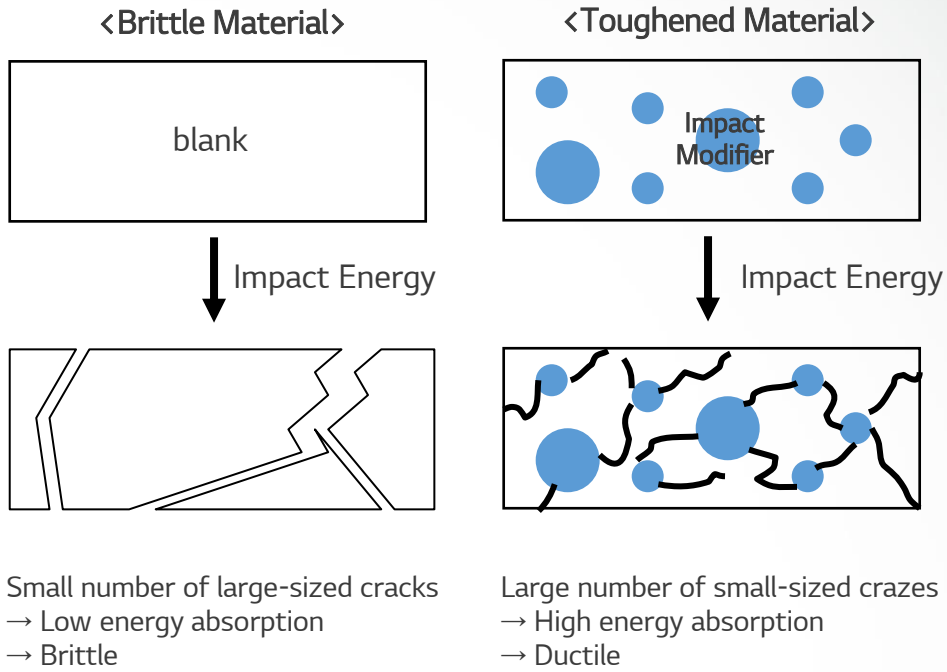
Core : Cross-linked Rubber

To enhance impact strength

Shell : Thermoplastic

To improve compatibility

Mechanism of Impact modifier



Small number of large-sized cracks
→ Low energy absorption
→ Brittle

Large number of small-sized crazes
→ High energy absorption
→ Ductile

Microstructures

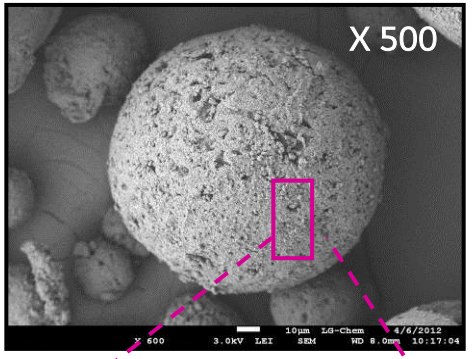
A powder of impact modifier consists of about 1 billion particles. This powder should be well-dispersed during processing to show good physical properties such as impact strength and so on.

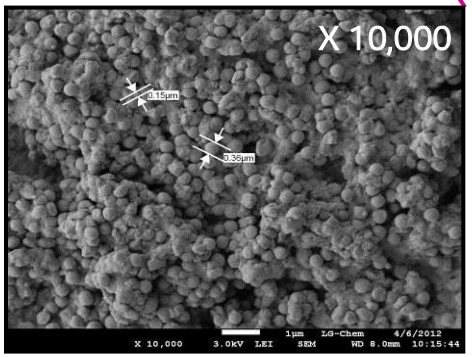
Powder



150um

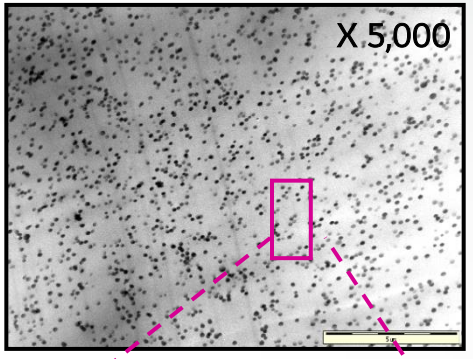
SEM (Powder)

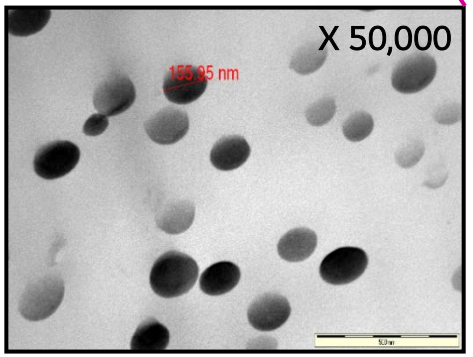




Processing

TEM (in Matrix)

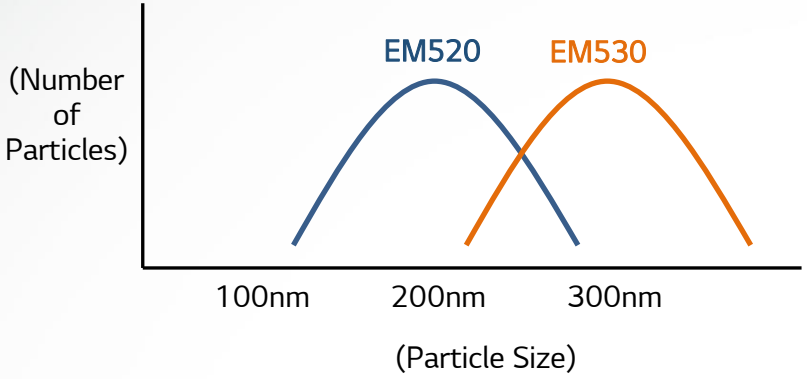




Characteristics - Particle Size and Compositions

EM520 and EM530 have different particle size and composition for their own properties.

Particle Size Distributions



- 200nm
  - Good for impact strength at low temperature
  - Minimization usage of emulsifier and mineral
- 300nm
  - Good for impact strength at room temperature
  - Improvement melt flow rate

Compositions of Impact Modifier

	Butadiene	Acrylate	Styrene	P/Size(nm)
LG EM520	O	O	-	~200
LG EM530	O	O	O	~300
Competitor D	73%	23%	4%	~200
Competitor K	71%	24%	5%	100~400

- EM520 and EM530 have optimized monomer composition for improving dispersibility and low TVOC.

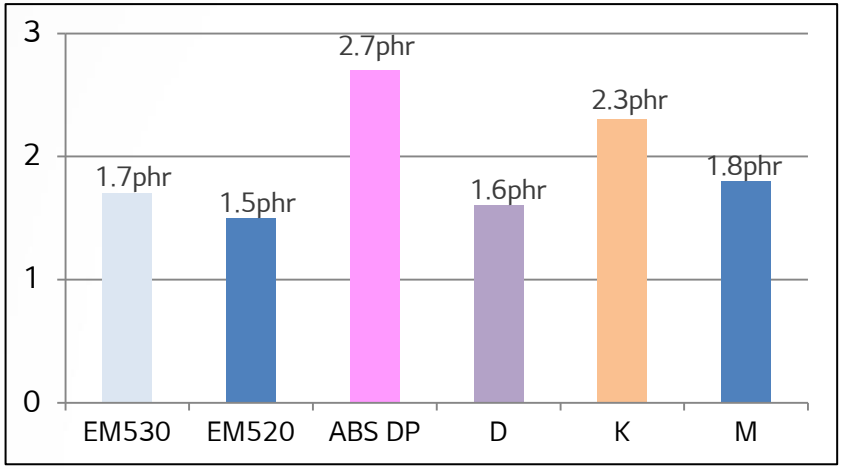
Characteristics – Mineral and Soap Contents

EM520 and EM530 minimizes mineral and emulsifier contents to improve thermal properties.

Mineral Contents

Grade	Ash Content	Component(EDS)
LG EM530	0.11%	Ca, Na, K, S
LG EM520	0.04%	Na, K
Competitor D	0.29%	P, Na, K
Competitor K	0.07%	Cl, Na
Competitor M	0.05%	Si, Na, K

Emulsifier Contents



Heat Stability Staying in Injection Molding (330℃, 10min)

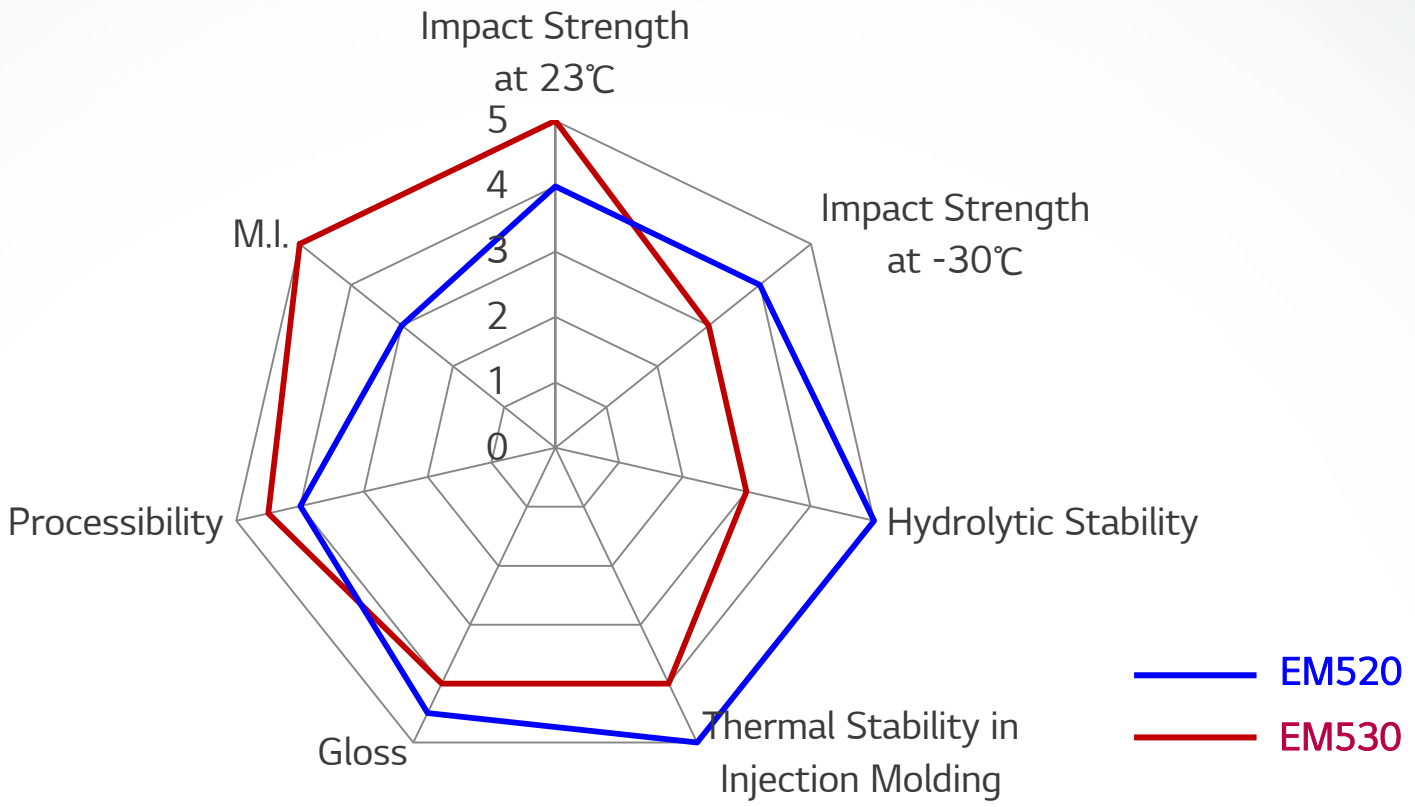
Before Staying

After Staying



Characteristics – EM520 vs. EM530

EM530 is suitable for large-area and complex products requiring thermal stability and EM520 is good for applications requiring hydrolytic stability.



Characteristics – LG vs. Competitors

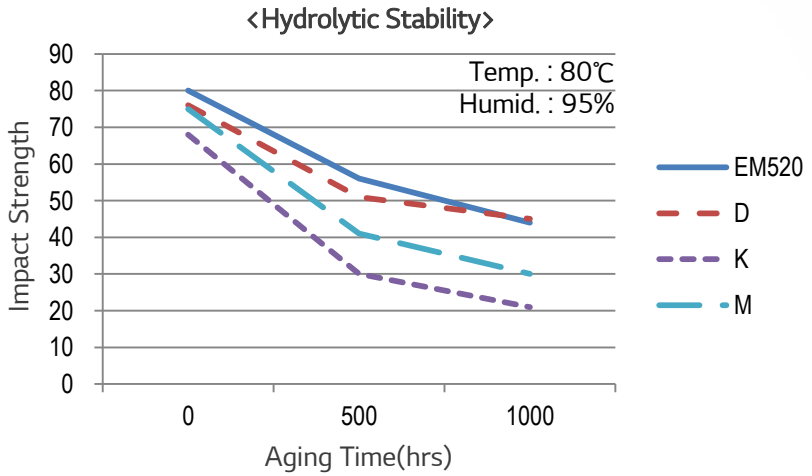
Grade	LG Chem	Kaneka	MRC
	EM530	M701	C223A
Impact Strength (23℃)	●●●●●	●●●●	●●●●●
Thermal Stability	●●●●	●	●●●●
Melt flow index	●●●●●	●●●	●●●●
Impact Strength (-30℃)	●●●	●●●	●●●
Hydrolytic Stability	●●●	●●●	●●
TVOC, Gas	●●●●	●●●	●●●
Coloring	●●●●	●●●●	●●●●

Grade	LG Chem	Kaneka		MRC	DOW
	EM520	M711	M732	C223A	EXL2690 EXL2620
Hydrolytic Stability	●●●●●	●●●	●●●●	●●	●●●●●
Thermal Stability	●●●●●	●●●●●	●●●●●	●●●●	●●●●●
TVOC, Gas	●●●●●	●●●●●	●●●●●	●●●	●●●●
Impact Strength (23℃)	●●●●	●●●	●●●	●●●●●	●●●●
Impact Strength (-30℃)	●●●●	●●●	●●●	●●●	●●●●
Melt flow index	●●●	●●●	●●	●●●●	●●
Coloring	●●●	●●●	●●●●●	●●●●	●●●●

●●●● Excellent ⟷ ● Poor

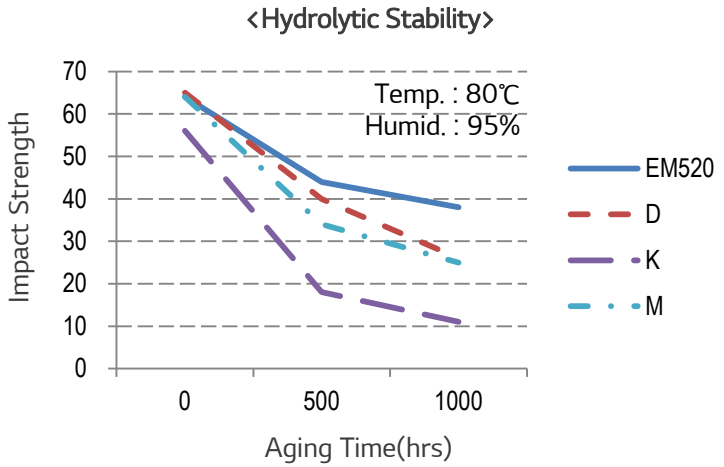
EM520 - Test Results in Flame Retardant PC Compound

Formulation : PC/F.R./MBS = 96/1/3%, White Color

		EM520	Competitor D	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	79.9	75.6	67.8	74.9
	Izod 1/8" (kgf.cm/cm), -30℃	29.7	28.1	24.3	26.5
	Tensile Strength (kgf/cm^2)	634	604	593	619
	Flexural Modulus (kgf/cm^2)	21,532	20,954	19,796	19,690
	M.I. (g/10min)	17.2	16.6	17.1	17.6
Optical Properties	Gloss (45°)	99	101	102	102
	TVOCs (ppm)	**320**	400	450	450
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Micro-crack	Micro-crack	Micro-crack
Thermal Stability					

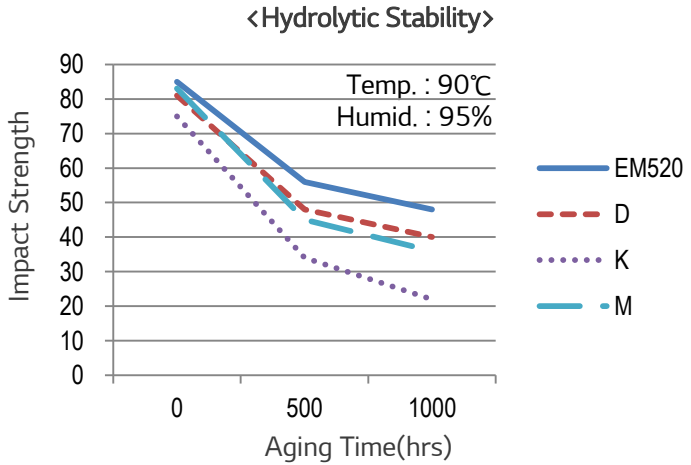
EM520 - Test Results in Flame Retardant PC/ABS Compound

Formulation : PC/F.R./ABS/MBS = 63/15/14/6%, Black Color

		EM520	Competitor D	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	63.9	64.8	56.2	64.4
	Izod 1/8" (kgf.cm/cm), -30℃	11.0	10.8	8.7	10.9
	Tensile Strength (kgf/cm^2)	480	481	475	483
	Flexural Modulus (kgf/cm^2)	23,156	22,374	23,334	22,914
	M.I. (g/10min)	33.5	29.3	31.7	31.4
Optical Properties	Gloss (45°)	94	89	98	93
	TVOCs (ppm)	**550**	910	1,000	930
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Micro-crack	Break	Micro-crack
Thermal Stability					

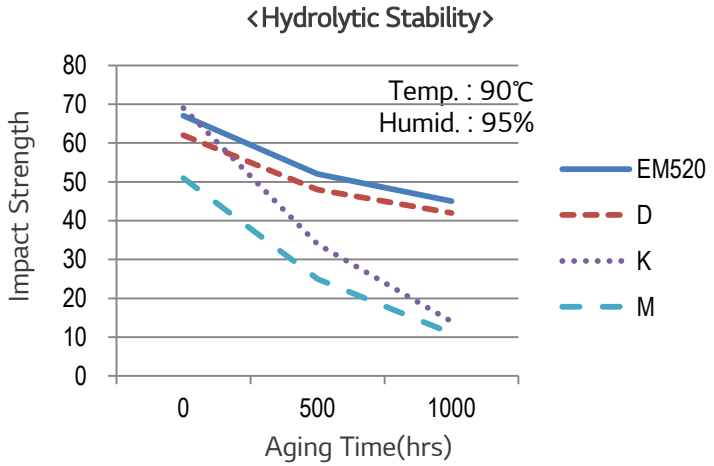
EM520 - Test Results in PC/PBT Compound

Formulation : PC/PBT/MBS = 63/30/5%, Black Color

		EM520	Competitor D	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	84.4	81.2	75.8	83.1
	Izod 1/8" (kgf.cm/cm), -30℃	23.3	18.9	20.7	19.0
	Tensile Strength (kgf/cm^2)	508	483	468	442
	Flexural Modulus (kgf/cm^2)	20,451	20,450	20,779	20,430
	M.I. (g/10min)	22.3	23.2	25.0	25.1
Optical Properties	Gloss (45°)	84	88	103	70
	TVOCs (ppm)	**350**	450	590	520
Chemical Resistance (R 1.3, Acetic acid 99.0%)		No Crack	No Crack	No Crack	No Crack
Thermal Stability					

EM520 - Test Results in PC/ASA Compound

Formulation : PC/ASA/MBS = 80/8/3%, Black Color

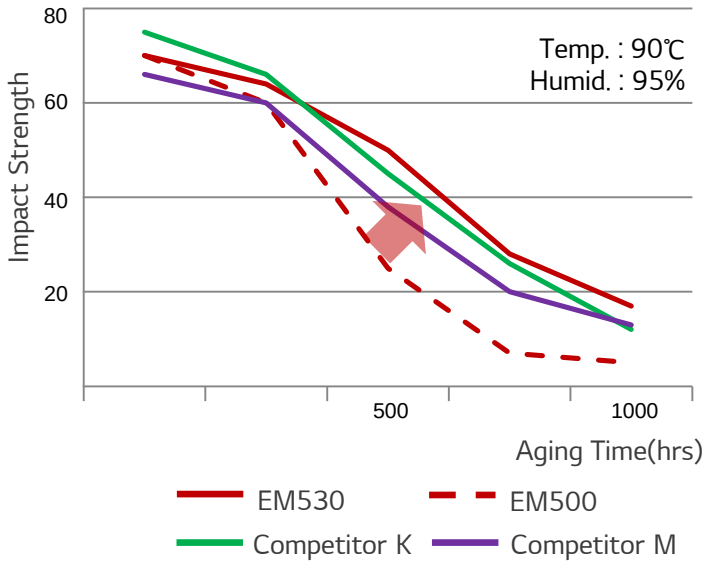
		EM520	Competitor D	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	67.2	62.3	69.6	51.0
	Izod 1/8" (kgf.cm/cm), -30℃	44.7	39.7	48.1	34.8
	Tensile Strength (kgf/cm^2)	523	590	623	618
	Flexural Modulus (kgf/cm^2)	19,582	19,527	19,907	19,200
	M.I. (g/10min)	18.8	18.4	16.7	15.1
Optical Properties	Gloss (45°)	14.1	11.3	13.8	9.8
	TVOCs (ppm)	**410**	510	1,300	1,200
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	No Crack	Break	Micro-crack
Thermal Stability					

EM530 - Test Results in PC Compound

Formulation : PC/MBS = 96/4%, Natural Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	72	70	69	72
	Izod 1/8" (kgf.cm/cm), -30℃	37	29	25	33
	M.I. (g/10min)	17.6	17.1	16.6	17.0
Optical Properties	Gloss (45°)	99	93	94	93
	Color (YI)	-3.5	-3.6	-4.1	-3.8
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Micro-crack	Micro-crack	Micro-crack
Thermal Stability					

<Hydrolytic Stability>



<Staying in Injection Molding>



Test Condition :330℃, 10min

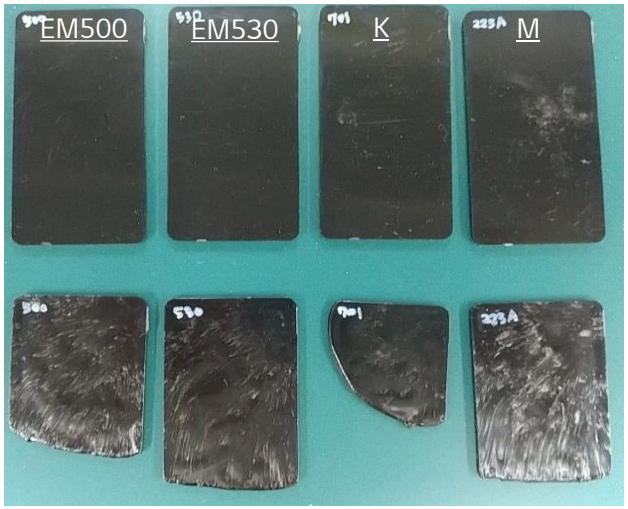
EM530 - Test Results in Flame Retardant PC Compound

Formulation : PC/F.R./MBS = 75/15/7%, Black Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	53	52	51	53
	Izod 1/8" (kgf.cm/cm), -30℃	18	13	14	17
	M.I. (g/10min)	37.7	35.5	34.9	36.7
Optical Properties	Gloss (45°)	92	89	81	89
	Color (L)	28.2	28.3	28.4	28.2
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Micro-crack	Micro-crack	Micro-crack
Thermal Stability		<Staying in Injection Molding>			

Before Staying

After Staying



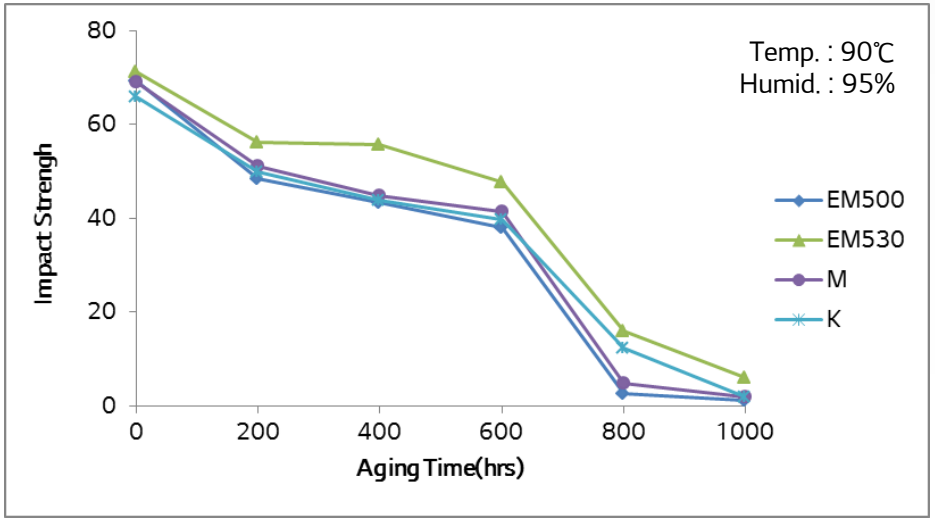
Test Condtion :300℃, 10min

EM530 - Test Results in PC/ABS Compound

Formulation : PC/ABS/MBS = 70/25/5%, Natural Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	71	69	66	69
	Izod 1/8" (kgf.cm/cm), -30℃	26	25	21	24
Optical Properties	Gloss (45°)	102	97	93	102
	Color (L)	86	87	89	87
Thermal Stability					

<Hydrolytic Stability>



EM530 - Test Results in Flame Retardant PC/ABS Compound

Formulation : PC/F.R./ABS/MBS = 64/15/13/6%, Black Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	62	56	60	61
	Izod 1/8" (kgf.cm/cm), -30℃	11	8	9	11
Optical Properties	Gloss (45°)	88	84	53	88
	Color (L)	27.9	27.8	27.6	27.8
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Crack	Micro-crack	Micro-crack
Thermal Stability					

<Staying in Injection Molding>



Test Condtion :280℃, 10min

EM530 - Test Results in Flame Retardant PC/SAN Compound

Formulation : PC/F.R./SAN/MBS = 63/14/14/9%, Natural Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	56	56	49	54
	Izod 1/8" (kgf.cm/cm), -30℃	12	10	12	11
	M.I. (g/10min)	16.4	16.2	14.0	16.0
Optical Properties	Gloss (45°)	88	86	71	88
	Color (YI)	9.7	9.9	7.1	8.4
Chemical Resistance (R 1.3, Acetic acid 99.0%)		Micro-crack	Micro-crack	Micro-crack	Micro-crack
Thermal Stability		<Staying in Injection Molding>			

Before Staying

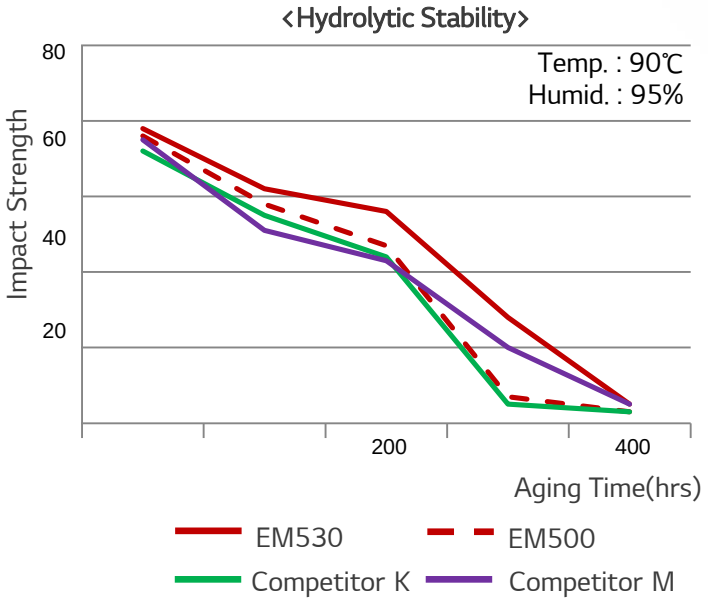
After Staying



Test Condtion : 280℃, 10min

EM530 - Test Results in PC/PBT Compound

Formulation : PC/PBT/MBS = 60/30/10%, Natural Color

		EM530	EM500	Competitor K	Competitor M
Physical Properties	Izod 1/8" (kgf.cm/cm), 23℃	76	78	72	73
	Izod 1/8" (kgf.cm/cm), -30℃	48	50	43	50
	M.I. (g/10min)	25.0	23.2	24.0	25.1
Optical Properties	Gloss (45°)	100	88	87	78
	Color (L)	25.5	25.2	25.0	25.1
Chemical Resistance (R 1.3, Acetic acid 99.0%)		No Crack	Micro-crack	No Crack	No Crack
Thermal Stability					



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